MIL-B-24001A
12 December 1966
SUPERSEDING
MIL-B-24001(DOCKS)
14 January 1964

MILITARY SPECIFICATION

BROILERS, ELECTRIC AND GAS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric and gas-fired broilers.

1.2 Classification. The broilers shall be of the following types, sizes, and grades as specified (see 6.2).

Type I - Electric
  Size 1 - Single deck (section)
  Size 2 - Double deck.

Type II - Gas
  Size 1 - Single deck

Grade A - Corrosion-resistant steel exterior
Grade B - Carbon steel, with black enamel finish exterior

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-766 - Steel Plates, Sheets and Strip, Corrosion Resisting

FSC 7310

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Military

MIL-R-11337 - Ranges, Packaging and Packing of

STANDARDS

Federal

Fed. Std. 66 - Steel: Chemical Composition and Hardenability

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Gas Association, Inc. (A.G.A.)

Directory of Approved Appliances and Listed Accessories

(Application for copies should be addressed to the American Gas Association, Inc., 1032 E. 62nd St., Cleveland, Ohio 44122.)

American Trucking Associations, Inc.

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P St., N.W., Washington, D.C. 20036.)

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National Bureau of Standards

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

National Sanitation Foundation (NSF)

Standard No. 4 - Electric Commercial Cooking and Warming Equipment
Standard No. 4 - Gas Commercial Cooking and Warming Equipment
Seal of Approval Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, School of Public Health, University of Michigan, Ann Arbor, Michigan 48103.)

Underwriters' Laboratories, Inc. (UL)

UL-197 - Commercial Electric Cooking Appliances
UL Electrical Appliance and Utilization Equipment List

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Ill. 60611; 1655 Scott Blvd., Santa Clara, California 95050; 333 Pfingsten Road, Box 247, Northbrook, Ill. 60062; or 1285 Walt Whitman Rd., Melville, L.I., New York 11749.)

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

United States of America Standards Institute (USASI)

Z21.3-1962 - Approval Requirements for Hotel and Restaurant Gas Ranges and Unit Broilers
Z21.18-1956- Listing Requirements for Domestic Gas Appliance Pressure Regulators

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th St., New York, N.Y. 10016.)

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Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), the contractor shall furnish, within the time limit specified in the contract or order, a preproduction sample for inspection. This inspection, which shall consist of the examination and tests specified in 4.5 and 4.6, shall be conducted prior to the submission for acceptance of the remainder of the unit furnished under the contract. The preproduction sample may be a special preproduction item or a standard or modified production line unit which meets the requirements of this specification. In any case, the preproduction sample shall be identical with the item the contractor proposes to furnish in fulfillment of the contract except that when the contract includes broilers which will vary on the basis of classification or options specified herein, the preproduction sample shall be designated by the contracting officer. Following acceptance by the Government, the preproduction sample may be delivered, after required servicing and adjustment, as the last unit of supply under the contract. Approval of the preproduction sample on the basis of the specified inspection shall not relieve the contractor of his obligation to meet the requirements of the specification for the remaining units furnished under the contract.

3.2 Standards compliance. Prior to approval of the preproduction sample, if one is specified, or prior to approval of the first shipment, if a preproduction sample is not specified, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the broilers he proposes to furnish under this specification meet the requirements of the following applicable standards:

a. Type I - NSF Standard No. 4 and UL Standard No. 197.

b. Type II - NSF Standard No. 4 and USA Standard Z21.3.

3.2.1 Certification. Acceptable evidence of meeting the requirements of the specified standards will be as follows:

a. NSF - NSF Seal of Approval or inclusion of the equipment in the current NSF Seal of Approval Listing of Food Service Equipment.

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b. UL - UL label, a copy of the UL Listing Card, or inclusion of the equipment in the current UL Electrical Appliance and Utilization Equipment List.

c. USASI - Photostatic copy of the American Gas Association, Inc. (AGA) Certificate of Approval or a listing in the current AGA Directory of Approved Appliances and Listed Accessories.

3.2.2 Alternate certification. In lieu of the certification specified in 3.2.1, acceptable evidence of meeting the requirements of the NSF, UL, and USASI standards will be a certified test report from a nationally recognized independent testing laboratory, acceptable to the cognizant military medical authority, attesting that the equipment offered has been examined and tested and that it conforms to the requirements of the applicable standards.

3.3 Standard product. Except as specified herein, the broiler and its component parts shall be a regular commercial product of the manufacturer or his suppliers. The items furnished shall be new and unused.

3.4 Material. Material shall be as specified herein. Material not definitely specified shall be in accordance with the recommendations of NSF Standard No. 4. Material shall be free from defects that may affect the serviceability of the broilers.

3.4.1 Dissimilar metals. Contact between dissimilar metals likely to cause deterioration of parts by galvanic corrosion shall be avoided. Where such contact cannot be avoided, joints between dissimilar metals shall be protected against galvanic corrosion by plating, coating, insulation, gaskets, or other suitable means. Bolts, nuts, pins, screws, and other fastenings shall be of the same material as the metals joined or shall be cathodic to the metals joined.

3.4.2 Corrosion-resistant steel. Corrosion-resistant steel shall conform to class 302 or 304 of QQ-S-766.

3.4.3 Carbon steel. Carbon steel shall be hot or cold rolled steel, annealed, conforming to the composition indicated for any AISI steel number from C1005 to C1095 specified in Fed. Std. No. 66.

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3.5 Interchangeability. All broilers of the same classification and furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, and accessories. No deviations will be acceptable without prior written approval by the contracting officer.

3.6 Design. The design of the broilers and accessories shall permit accessibility for maintenance and service in the field. The broilers shall be designed to prevent conditions hazardous to personnel. The broiler exterior shall have a temperature that enables personnel to work in close proximity to the broiler without being burned. Ample protection against electrical shock shall be provided. Screw threads shall conform to the requirements of Handbook H28.

3.6.1 Type I. The electric broiler shall consist of an insulated container having adjustable grids, drip pans, electric heaters, junction box, flue assembly, temperature controls and other components required for satisfactory operation. The components shall be assembled and ready for use when connected to the appropriate electric power source and flue. The broiler shall consist of either one or two decks (sections). Each section shall be capable of operating independently or in combination. Electric broilers shall conform to the nominal dimensions of table I.

Table I. Nominal dimensions of electric broilers (inches ± 2 inches)

Size	1	2
Overall		
Height to flue top	57½ - 66	71 - 78
Width	36	36
Depth	36	36
Grid		
Width	25	25/Sect.
Depth	22	22/Sect.

3.6.2 Type II. The gas broiler shall consist of a stand-mounted, thickly insulated, open-front container having an adjustable grid, drip pans, flue assembly, a universal heater, temperature controls

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and other components required for satisfactory operation. The components shall be assembled and ready for use when connected to the fuel supply and flue. The broiler shall consist of one deck. Gas broilers shall conform with the nominal dimensions in table II.

Table II. Nominal dimensions of size 1 gas broilers (inches ± 2 inches)

Overall	
Height to flue top	61 - 66
Width	36
Depth	36
Grid	
Width	25 - 28
Depth	28

3.7 Capacity. The type I broiler shall be capable of providing prepared foods in the minimum quantity per hour as indicated below:

Food	Broiler size 1	2
Sirloin steak (8 oz.)	180	360
Chicken (broiler halves)	30	60

The capacity of type II broilers shall be adequate for the intended use.

3.8 Construction. Broilers shall be constructed in accordance with the requirements herein and the general construction provisions of NSF Standard No. 4. Where a difference may exist, this specification shall take precedence. The cabinet shall be of all-metal construction, and shall have adequate stability to withstand the stresses, jars, and vibrations incident to shipping, storage, and operation. The body for grade A broilers shall be fabricated from any class of corrosion-resistant steel specified in 3.4.2. The body for grade B shall be fabricated from carbon steel specified in 3.4.3.

3.8.1 Wall panels. Wall panels shall consist of double walls enclosing a suitable insulating material. Wall panels, except those corrosion-resistant or removable, shall be fitted together and sealed to provide a moisture-tight enclosure.

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3.8.2 Base. The base shall consist of double wall construction enclosing a suitable insulating material and in all other respects shall be in accordance with the manufacturer's standard practice.

3.8.3 Insulation. The insulation for the walls, top, and base shall be rock wool, glass wool, or other suitable insulation material. The insulation shall be a nonmoisture absorbing material, so installed as not to separate, compact, or settle under normal conditions of shipment and use. The insulation shall be such that with the broiler at a temperature of 450° F. the average temperature of the outside walls shall not exceed the ambient temperature by more than 25° F. Loose fill insulation is not acceptable.

3.8.4 Grids. The broiler shall be provided with sturdily constructed, corrosion-resistant steel grids in each of its sections. The grids shall be easily removable and shall be adjustable as to height by operation of a conveniently located handle. Grids shall be adjustable throughout a vertical range of 6 inches.

3.8.5 Drip pans. The broiler shall be furnished with a removable drip pan fitted in the bottom of a corrosion-resistant broiling compartment.

3.8.6 Flue assembly. Each broiler shall be provided with a vent flue at the rear so located as to be capable of connection to another section. The flue assembly shall be of sufficient size to remove all broiler fumes and to preclude the possibility of an explosion. Gas-type broilers shall be furnished with a minimum of one square foot of explosion vent for every 30 cubic feet of broiler volume.

3.9 Heating equipment.

3.9.1 Type I. The broiler shall be completely wired at the factory for the electrical characteristics specified herein. Unless otherwise specified (see 6.2), the broiler shall be operated on a nominal 208-volt, 60 cycle, single- or 3-phase a.c. system. The electric heating unit shall be capable of preheating the broiler to broiler temperature within a period of 12 minutes. The heating unit for each section shall be rated at a minimum of 12 kilowatts. The temperature shall be automatically regulated by an adjustable temperature control. The heater shall be constructed to afford adequate protection against accidental encounter.

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3.9.2 Type II. The fuel system, including burners, shall be designed to efficiently burn natural, mixed, or liquefied petroleum gas. Burner gas valves and pilot gas valves shall be of the fixed-orifice type with interchangeable orifice hoods. Each broiler section shall be provided with three orifice hoods, each drilled for one of the three types of gas specified above. The manufacturer shall provide instructions for conversion from any one type of gas to the other two. The valve hoods, shall be color-coded as follows: (1) Natural gas - unpainted brass, (2) mixed gas - red, and (3) liquefied petroleum gas - black. Burner supports shall be securely installed in correct position. The broiler shall be provided with a gas-appliance pressure regulator for the burner. Regulators shall conform to the requirements of USASI Z21.18, "Listing Requirements for Domestic Gas Pressure Regulators". The broiler shall have a minimum input of 76,000 B.t.u. per hour and shall preheat to broiling temperature within 12 minutes.

3.10 Finish. All exterior surfaces of corrosion-resistant steel broilers shall have a number 4 standard polish. Exterior surfaces of grade B broilers shall have a black lacquered finish in accordance with the manufacturer's standard practice. Broiler interiors shall be finished in accordance with the manufacturer's standard practice.

3.11 Instruction plates. Instruction plates indicating special procedures and precautions for operating or servicing the unit shall be permanently attached to each broiler.

3.12 Technical publications. Unless otherwise specified (see 6.2), one complete set of installation, operation, and maintenance instructions, including a replaceable parts list shall be furnished with each broiler.

3.13 Identification marking. The broiler shall be marked for identification in accordance with MIL-STD-130. The approved equipment nomenclature shall be: Broiler (Electric or Gas as applicable).

3.14 Workmanship. Standards of workmanship shall be as established by the applicable NSF, UL and USASI standards referenced herein and shall be of the quality prevailing among manufacturers normally producing broilers of the type specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used are manufactured and inspected in accordance with the specified requirements of referenced subsidiary specifications and standards or, if none, in accordance with this specification.

4.1.2 Standards compliance. The manufacturer shall make available to the contracting officer or his authorized representative evidence of compliance with referenced USASI, UL, and NSF standards. Certification specified in 3.2.1 or 3.2.2 will be accepted as evidence of compliance.

4.2 Inspection lot. All broilers of the same type, size, grade and options offered for delivery at one time shall be considered a lot for inspection purposes.

4.3 Inspection classification and requirements. Inspection shall be classified as follows:

a. Preproduction inspection.
b. Quality conformance inspection.

4.3.1 Preproduction inspection. Preproduction inspection shall be performed on the preproduction sample (see 3.1). Unless otherwise specified (see 6.2), this inspection shall include the examination described in 4.5 and all tests of 4.6.

4.3.2 Quality conformance inspection. Quality conformance inspection shall be performed on the samples selected in accordance with 4.4 from each inspection lot. This inspection shall include the examination of 4.5 and the test of 4.6.1.

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4.4 Sampling for lot acceptance.

4.4.1 Sampling for examination. A random sample of broilers shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105. The acceptable quality levels shall be 1.5 percent defective units for major defects and 4.0 percent defective units for minor defects.

4.4.2 Sampling for tests. A random sample of broilers shall be selected from each inspection lot in accordance with MIL-STD-105 at inspection level S-4. The AQL shall be 4.0 percent defective.

4.5 Examination. The broiler shall be visually and dimensionally examined to determine conformance with the requirements of this specification. Any broiler containing one or more defects shall be rejected.

4.6 Tests.

4.6.1 Functional test. The broiler shall be installed and operated for a period of one hour in accordance with the manufacturers' instruction to verify that all components will function in a manner that insures compliance with this specification. During the test all controls shall be operated to determine proper functioning.

4.6.2 Insulation test. Maintain the broiler at a temperature of 450° F. for a period of one hour. At the end of that time, the temperature immediately adjacent to the outside surface of the broiler shall not exceed the ambient temperature by more than 25° F.

4.6.3 Preheat time. The broiler shall preheat to broiling temperature from an ambient temperature of 75° F. within 12 minutes.

4.7 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the equipment shall be inspected to determine compliance with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each broiler shall be preserved and packaged in accordance with the applicable level A requirements of MIL-R-11337.

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5.1.2 Level C. Each broiler shall be preserved and packaged in accordance with supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Each broiler shall be packed in accordance with the applicable level A requirements of MIL-R-11337.

5.2.2 Level B. Each broiler shall be packed in accordance with the applicable level B requirements of MIL-R-11337.

5.2.3 Level C. The broiler shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules or with National Freight Classification Rules.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Broilers covered by this specification are intended for use in the preparation of food at naval shore establishments and at hospitals of the Navy, Army, and Air Force.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of specification.

(b) Type, size, and grade required (see 1.2).

(c) When a preproduction sample is not required (see 3.1).

(d) Electric power characteristics (voltage, phase, frequency), if different (see 3.9.1).

(e) Technical publications, if different (see 3.12).

(f) Preproduction inspection, if different (see 4.3.1).

(g) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

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6.3 Preproduction sample. When a contractor has been the supplier of identical equipment in recent orders, the contracting officer may waive the requirement for a preproduction sample in the contract or order, and arrange for an accompanying reduction in price. However, the invitation for bids should include the requirement for preproduction sample when the successful bidder is a manufacturer whose unit has not passed the preproduction test.

6.4 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - YD
Air Force - 84
Review activities:
Army - MD
Navy - MS
Air Force - 03, 84

Code "C"

Preparing activity:
Navy - YD

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Commanding Officer
U.S. Naval Construction Battalion Center
Code 310
Davisville, Rhode Island 02854

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-B-24001A: Broilers, Electric and Gas

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 IA 0101-807-2000 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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